



Earth System

Earth System Science

Chapter 3: Cycles Of Major Biogeochemical Elements



Objective

- *To Explain The Carbon Cycle*
- *To Know The Reservoirs of Carbon*
- *To See The Carbon Sequestration and Sinks*
- *To be aware of carbon trading*

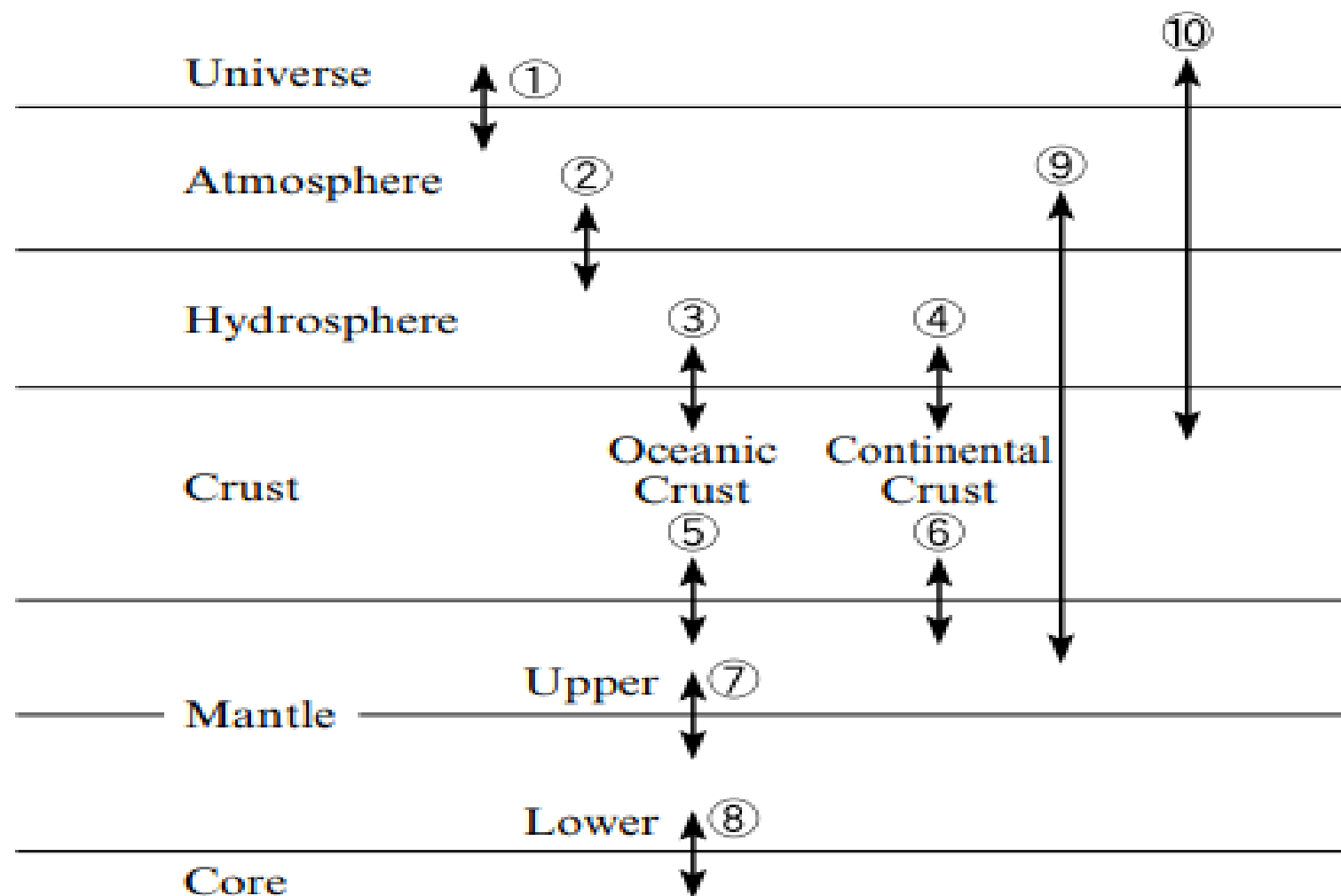
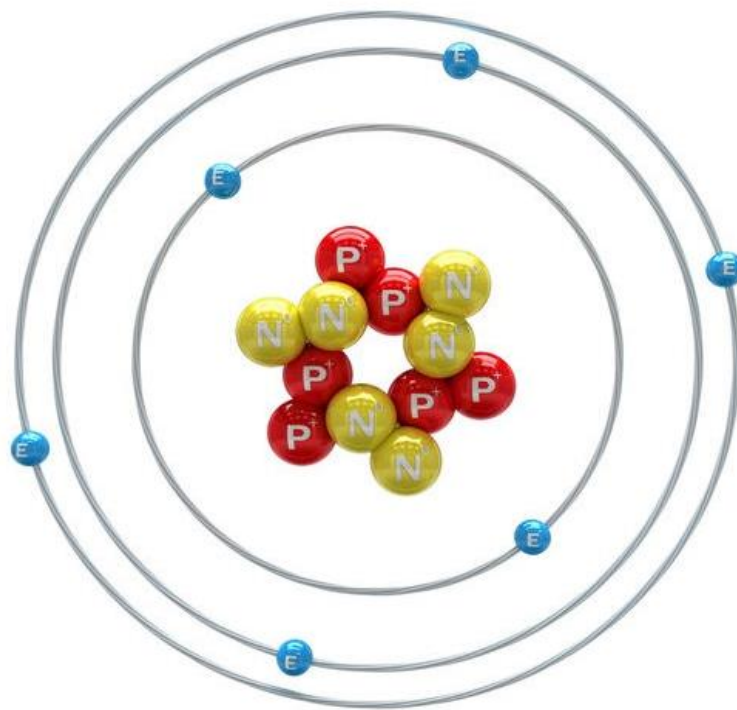


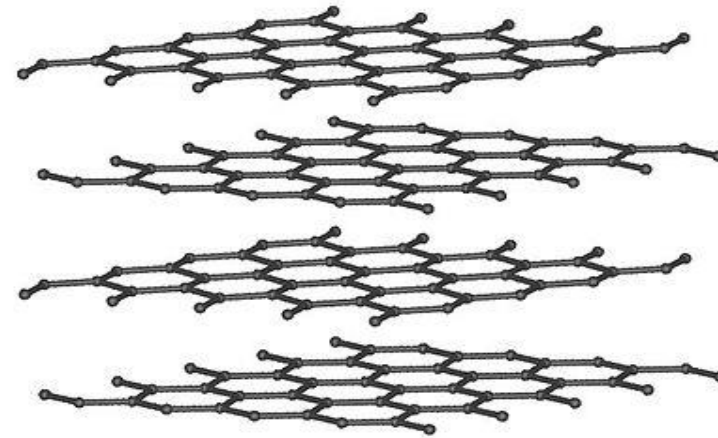
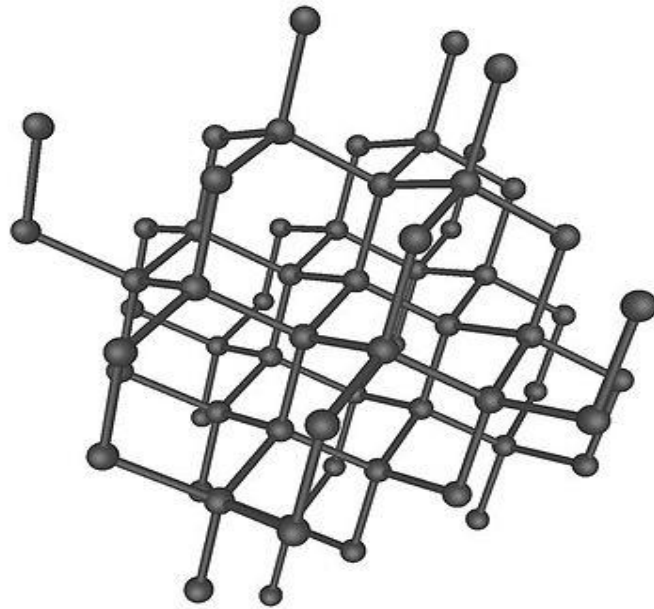
Fig. 3.1 The interactions in multispheres in the earth system. ① Formation of ozone layer. ② Dissolution of CO_2 . ③ Weathering of basalt, circulation of hydrothermal solution, formation of hydrothermal ore deposits, sedimentation. ④ Ground water–soil–rock interaction, sedimentation, formation of weathering ore deposits. ⑤ Subduction of plate, metamorphism, igneous activity. ⑥ Igneous activity. ⑦ Mantle convection. ⑧ Core–mantle interaction. ⑨ Volcanic gas, igneous activity. ⑩ Meteorite impact

The Carbon Cycle



Carbon?

- Carbon is the fourth most abundant element in the universe after hydrogen, helium, and oxygen.
- The three forms (allotropes) of carbon found freely in nature are the *amorphous, graphite, and diamond*.
- Graphite is one of the softest known materials, while diamond is the hardest known material.
- All forms of carbon are solids under normal temperatures and pressure conditions.
- Carbon is present in all known life forms.



Diamond and graphite are two allotropes of carbon: pure forms of the same element that differ in crystalline structure.

Reservoirs of Carbon

- Carbon cycle refers to the flow of carbon among the various global “reservoirs” that store carbon.
- Thus understanding the carbon cycle requires distinguishing the carbon reservoirs
- More than 99.9% of carbon is stored in rock reservoirs, which nearly accounts 50×10^{15} metric tons of carbon.

1 metric ton = 1000kg

- All the other reservoirs account only 42×10^{12} metric tons of carbon.
- These reservoirs are:
 - ✓ The atmosphere, where carbon exists mainly as CO_2
 - ✓ Soil, where carbon occurs in a variety of organic compounds; also as CO_2
 - ✓ The Ocean, where it is present mainly as dissolved bicarbonate ion (HCO_3^-)
 - ✓ Living Biomass
- Of the above carbon reservoirs, the ocean is the largest carbon stock, which holds 39×10^{12} metric tons of carbon

Sizes of carbon reservoirs

		metric tons
Rock Reservoir		50×10^{15}
	Limestone (mainly made of CaCO_3)	40×10^{15}
	Organic carbon in sedimentary rocks	10×10^{15}
	Fossil fuels	4.7×10^{15}
	Marine carbonate sediments	2.5×10^{15}
The Ocean		39×10^{12}
	Bicarbonate ion (HCO_3^-)	37×10^{12}
	Carbonate ion	1.3×10^{12}
	Dissolved CO_2	0.74×10^{12}
Organic carbon in soils and terrestrial sediments		1.6×10^{12}
Organic carbon in permafrost		0.9×10^{12}
Atmospheric CO_2		0.76×10^{12}
Living biomass		0.6×10^{12}

- It is convenient to see carbon cycle as two cycles rather than one
 - I. The long-term carbon cycle:
which requires hundreds of thousands to million of years to operate
 - II. The short-term carbon cycle:
which operates in timescales measured only in months to years to centuries.
- It is the short-term cycle that dictate the present climate and the way it changes over human time scales.

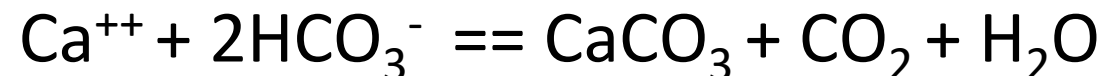
The long term carbon cycle

- This involves the movement of carbon between the solid Earth and the ocean and the atmosphere
- It starts with weathering of rocks, a process which removes CO_2 from the atmosphere
- This happens because atmospheric CO_2 reacts with water and with the silicate and carbonate minerals of rocks.
- This results in the formation of
 - ✓ Ions of calcium and magnesium
 - ✓ Bicarbonate (HCO_3^-)
 - ✓ Silica (SiO_3 , SiO_4)

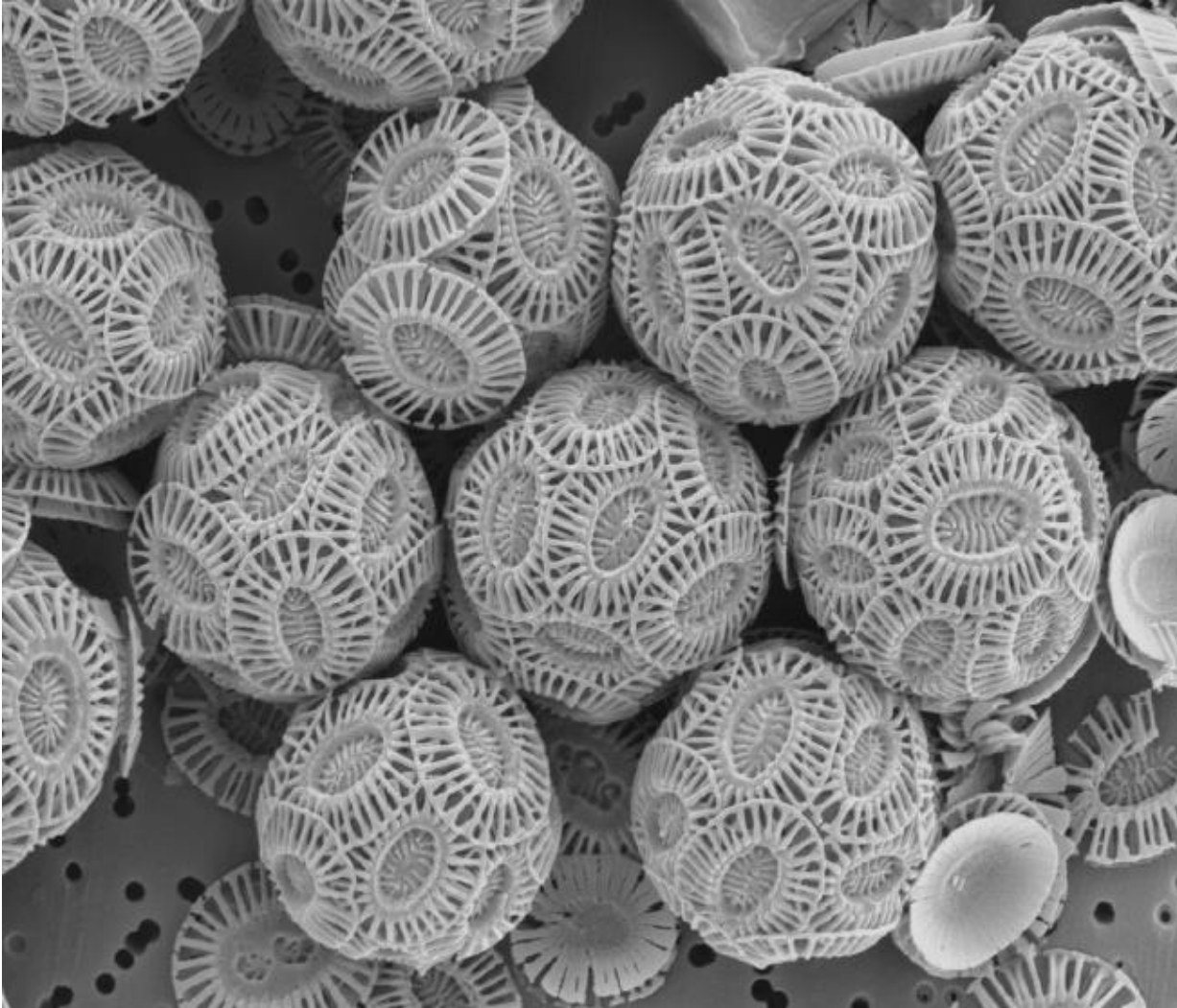
in solution in the water

- With weathering, atmospheric CO₂ reacts with water and with the silicate and carbonate minerals of rocks
- The simplified reaction can be written as:

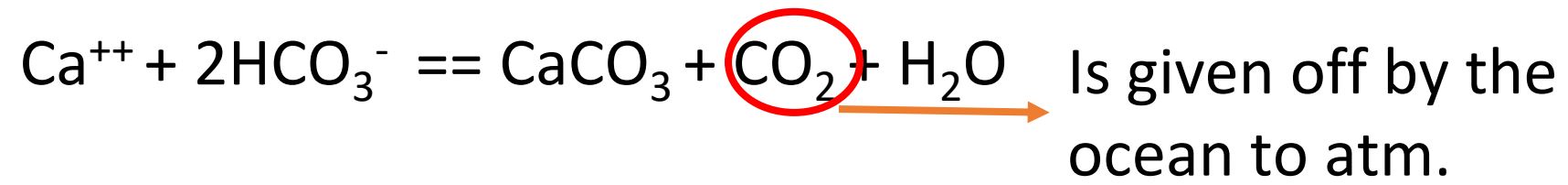
$$\text{CO}_2 + \text{H}_2\text{O} + \text{CaSiO}_3 + \text{MgSiO}_3 \rightleftharpoons \text{Ca}^{++} + \text{Mg}^{++} + 4\text{HCO}_3^- + 2\text{H}_4\text{SiO}_4 \text{ (silic acid)}$$
- The dissolved ions then wash into rivers and eventually end up in the ocean
- In the oceans organisms usually use the calcium and bicarbonate ions to make shells



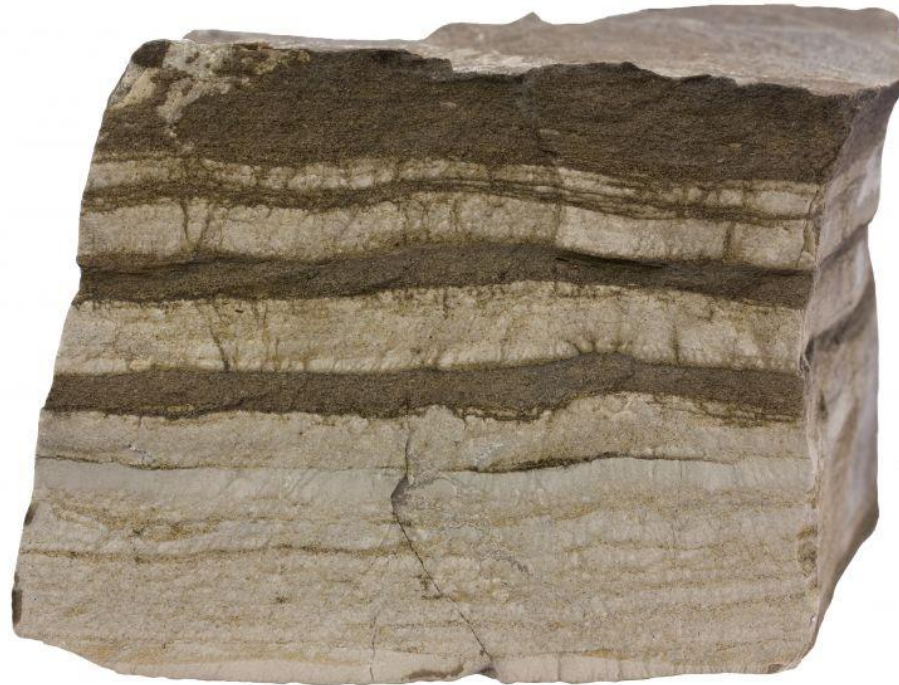
- Example: An organism, called *Emilania huxleyi*, makes delicate CaCO_3 shell as covering.



- It lives near the ocean surface where it photosynthesis

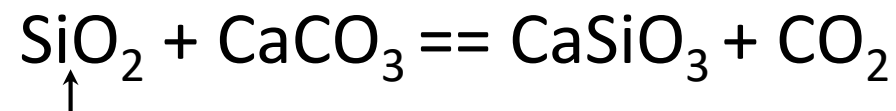


- As shelled organisms die, their carbonate shells accumulate on the ocean floor.
- New layers of sediment cover older layers, and with burial the sediments slowly transform to form **limestone**.



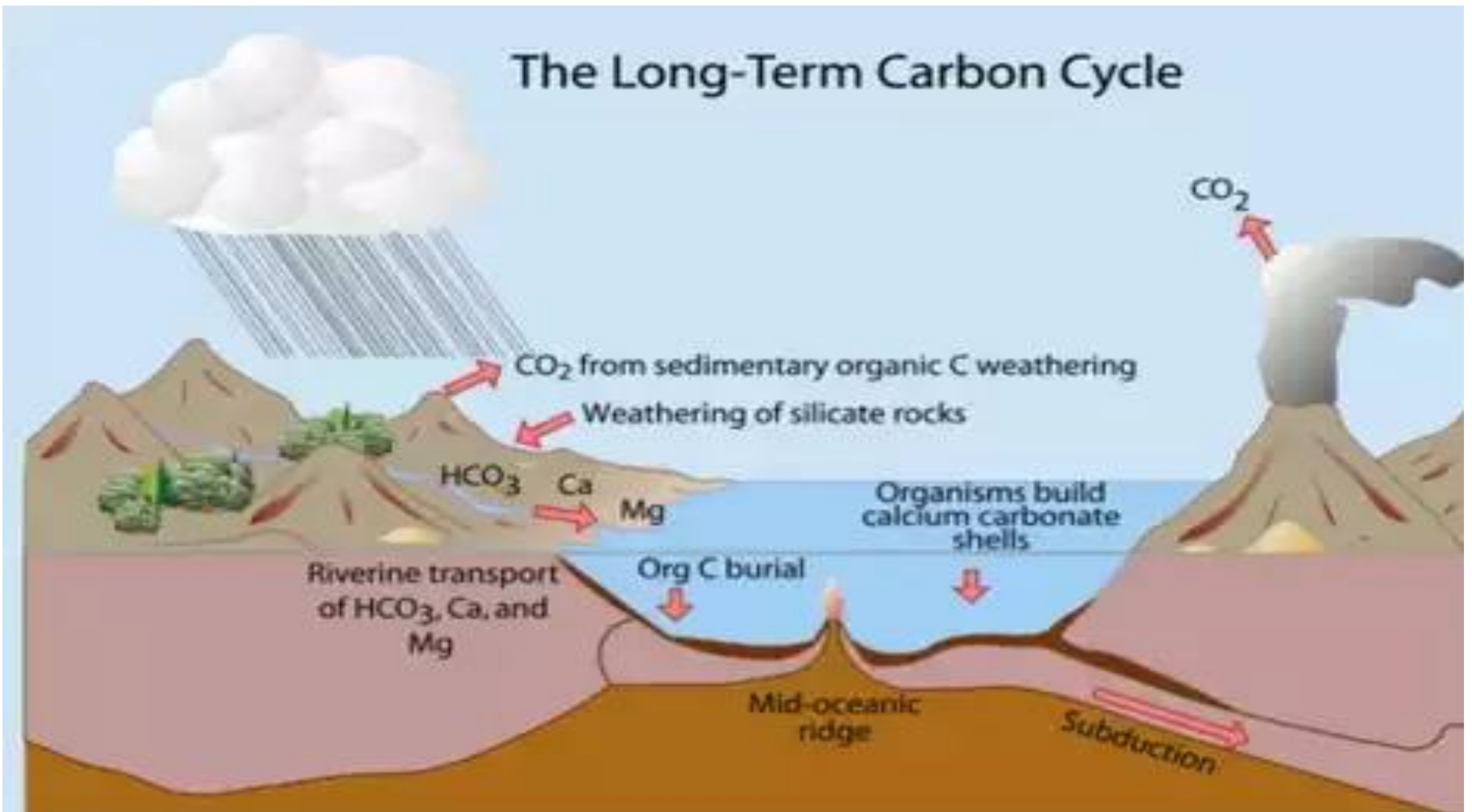
- The rocks are sometimes buried to depths of thousands of meters, in millions of years timescale
- In this case, the *heat and pressure* of the deep Earth may break down the carbonate minerals to give off CO₂
- The CO₂ then slowly percolates out of the crust and back into the atmosphere, completing the cycle

- Alternatively, the sedimentary rocks may be dragged down into earth's mantle in subduction zone.
Subduction zone: is a zone where one Earth's plate is thrust beneath another
- Here, the carbonates similarly break down and the CO₂ escapes to surface when a volcano erupts
- The reaction that takes place during the break down, in either case, is



↑
Silicon dioxide
(Silica) (Quartz)

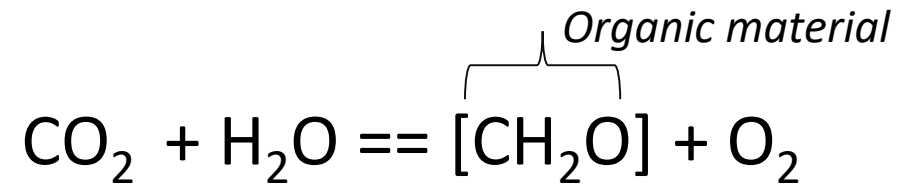
The Long-Term Carbon Cycle



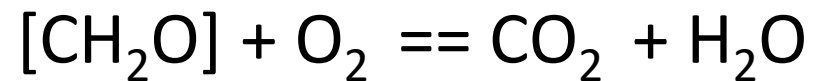
The short term carbon cycle

- This refers to the circulation of carbon among the ocean, atmosphere, soil and biosphere.
- It does not involve rocks at all
- The cycling is rapid: months to decades to millennium
- Hence, the short-term carbon cycle is the cycle that determines the fate of the climate over the next several decades.

- It starts with photosynthesis, a process that removes CO₂ from the atmosphere:



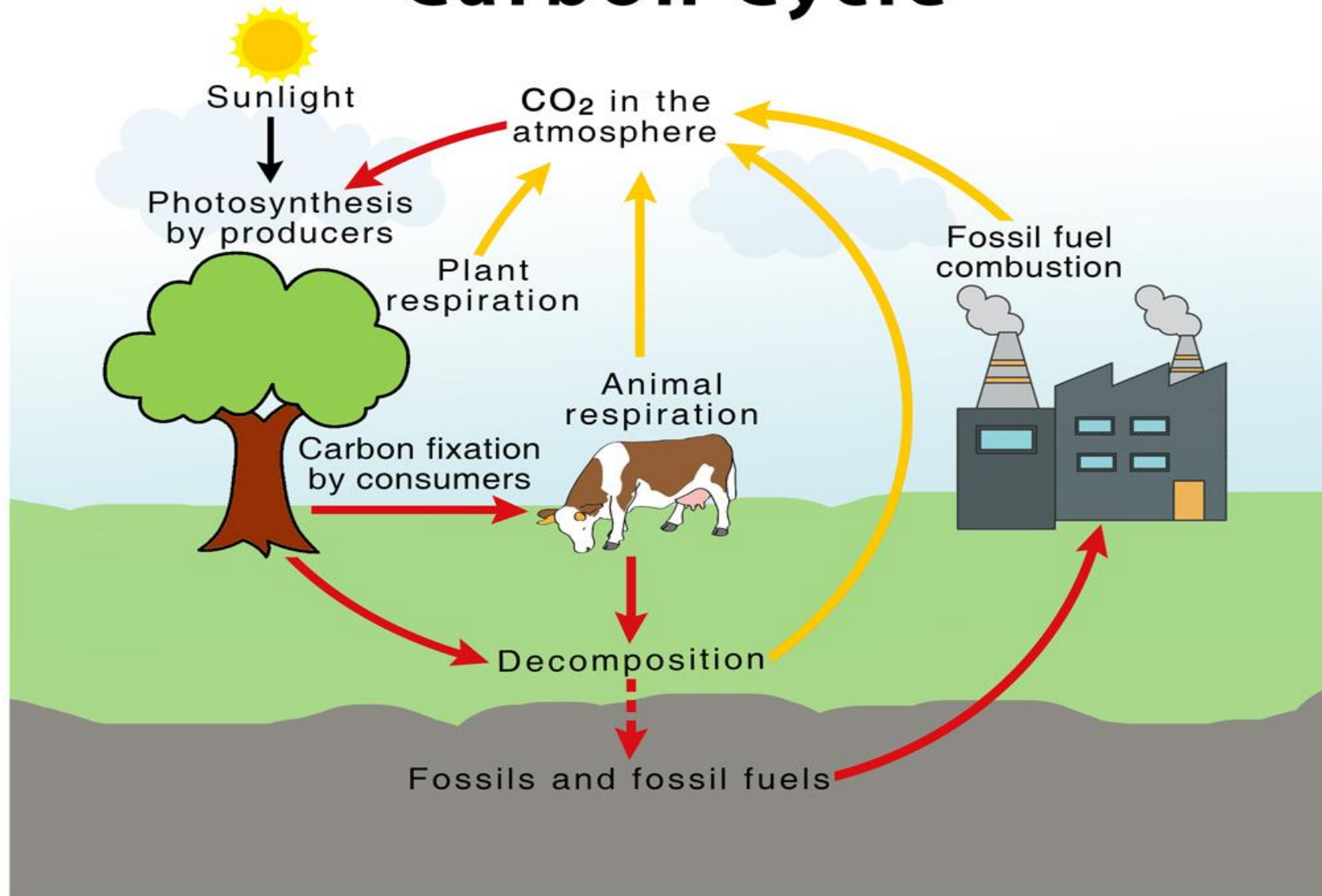
- About half of the terrestrial organic material from photosynthesis is returned to the atmosphere by respiration of plants and animals, including humans:



- The remainder ends up in soil by accumulation of dead debris.
- Then, microbes in the soil carry out aerobic respiration on dead matter to produce CO₂.
- Sometimes, if the microbes live in deeper soil levels where there is poor oxygen, they produce methane (CH₄) from the organic material via anaerobic respiration
$$2[\text{CH}_2\text{O}] \Rightarrow \text{CO}_2 + \text{CH}_4$$
- And the methane enters the atmosphere but eventually combines with oxygen to form more CO₂

- The burning of fossil fuels represents a net transfer of carbon from the long term rock reservoirs to short term surface reservoirs.
- This transfer is irreversible on timescales relevant to humanity.
- By burning fossil fuel, humans are removing carbon from the long term reservoirs and putting it into the short term reservoirs.
- *The shortest period of time that the long term carbon cycle requires to correct sudden perturbations in climate system may range 40,000 to 60,000 years.*

Carbon Cycle



Why the Carbon Cycle Matters?

- ❑ Regulates Earth's temperature through the greenhouse effect
- ❑ Supports photosynthesis and respiration (basis of life)
- ❑ Links biological, chemical, and geological processes

Carbon Sinks Key Players in the Global Carbon Cycle

- ❑ A carbon sink is any reservoir (natural or artificial) that absorbs more carbon than it releases.
- ❑ Helps mitigate climate change by reducing CO₂ in the atmosphere.
- ❑ It plays important role to balance carbon sources (emissions) and carbon sinks (absorption).
- ❑ Focus is on where carbon is stored in the Earth system.

Examples: Forests, oceans, soil

Types of Carbon Sinks

1. Terrestrial Carbon Sinks

- Forests
- Soil

2. Oceanic Carbon Sinks

- Physical pump:
- Biological pump

3. Engineered Carbon Sinks

- Carbon Capture and Storage (CCS)
- Biochar
- Direct air capture

Importance of Carbon Sinks

- Mitigate climate change by reducing atmospheric CO₂
- Maintain ecosystem balance and soil fertility
- Support ocean and terrestrial biodiversity

Threats to Carbon Sinks

- Deforestation → reduces terrestrial sinks.
- Ocean acidification → reduces CO₂ absorption.
- Soil degradation → releases stored carbon.
- Climate change → fires, releases carbon

Carbon Sequestration

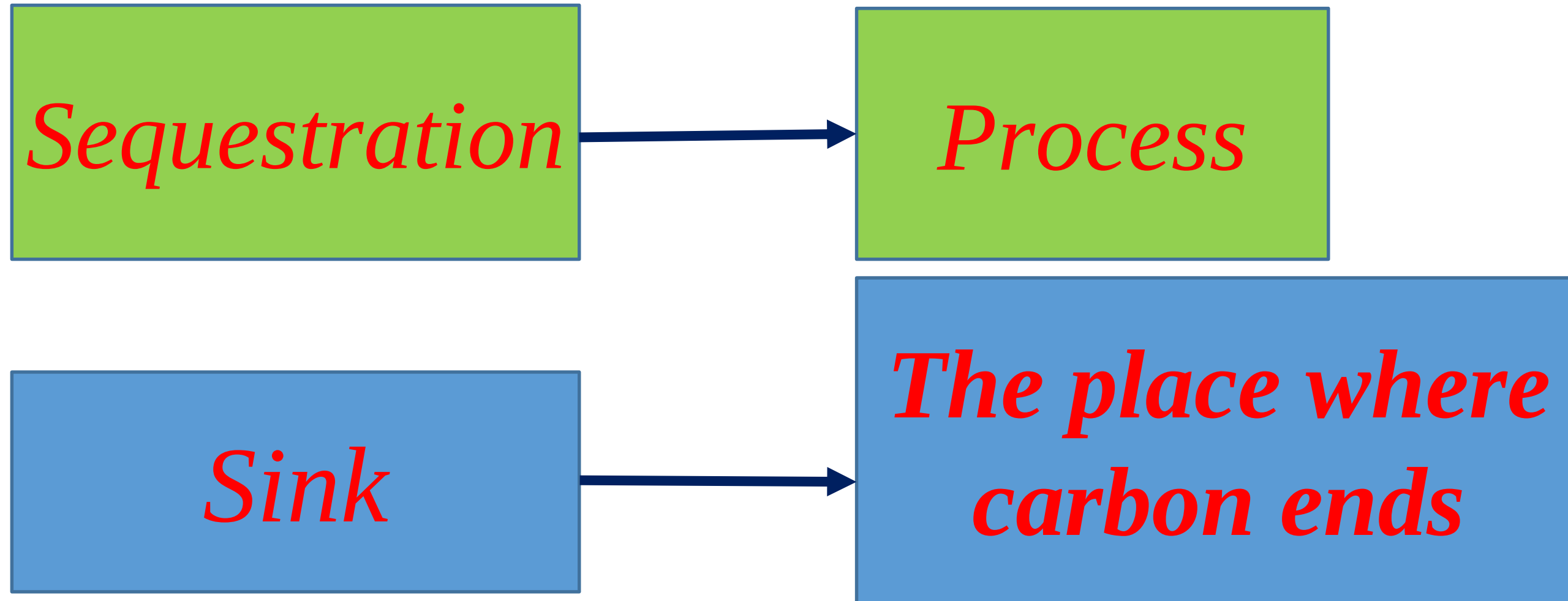
- ❑ Carbon sequestration is *the process* of capturing and storing carbon dioxide (CO₂) from the atmosphere to reduce global warming.
- ❑ Focus is on the action or method of moving carbon into a sink.
- ❑ It's essentially nature or technology

Examples: Trees capturing CO₂ via photosynthesis

Injecting CO₂ Underground (CCS)

Adding biochar to soil.

Carbon Sequestration Vs. Sink



Carbon Trading

- ❑ Carbon trading is a market-based approach to controlling greenhouse gas emissions.
- ❑ Companies or countries can buy or sell carbon credits to meet emission reduction targets.
- ❑ It aims to Incentivize for reducing emissions efficiently

Why Carbon Trading?

- Climate change requires reducing CO₂ emissions
- Not all emitters can reduce emissions at the same cost
- Carbon trading allows flexible, cost-effective reductions

Cont....

- ❑ Carbon Credit is A permit allowing one ton of CO₂ (or equivalent GHG) to be emitted
- ❑ Carbon Offset: Reduction in emissions elsewhere to compensate for emissions
- ❑ Carbon Market: Platform for buying and selling credits.

How Carbon Trading Works

- Government/authority sets an emission cap for sectors or companies
- Companies receive or buy allowances (1 allowance = 1 ton CO₂)
- If emissions < allowances → sell excess credits.
- If emissions > allowances → buy additional

Types of Carbon Trading

- *Compliance Markets:* Legally mandated, e.g., EU Emissions Trading System (EU ETS)
- *Voluntary Markets:* Companies voluntarily reduce emissions for reputation.

Global Examples

- ❑ EU ETS: Largest cap-and-trade system in the world
- ❑ California Cap-and-Trade Program: Covers multiple sectors
- ❑ Kyoto Protocol & Clean Development Mechanism (CDM): International projects generating credits

Carbon Trading Benefits and Criticisms

Benefits of Carbon Trading

- Cost-effective way to reduce emissions.
- Encourages innovation in clean technologies.
- Promotes global cooperation.
- Provides economic incentives for reducing emissions.

Criticisms / Challenges

- Carbon markets can be manipulated or oversupplied, reducing effectiveness.
- May allow “pollute now, pay later” mentality.
- Monitoring and verification of emission reductions can be difficult.

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Thanks !

Earth System Science

Chapter 3: Cycles Of Major Biogeochemical Elements



Objective

- *To Explain The Nitrogen Cycle*
- *To Know The five processes that make up the nitrogen cycle*
- *To See Importance of the Nitrogen Cycle in Meteorology*
- *To be aware of The phosphorous cycle*

The Nitrogen Cycle

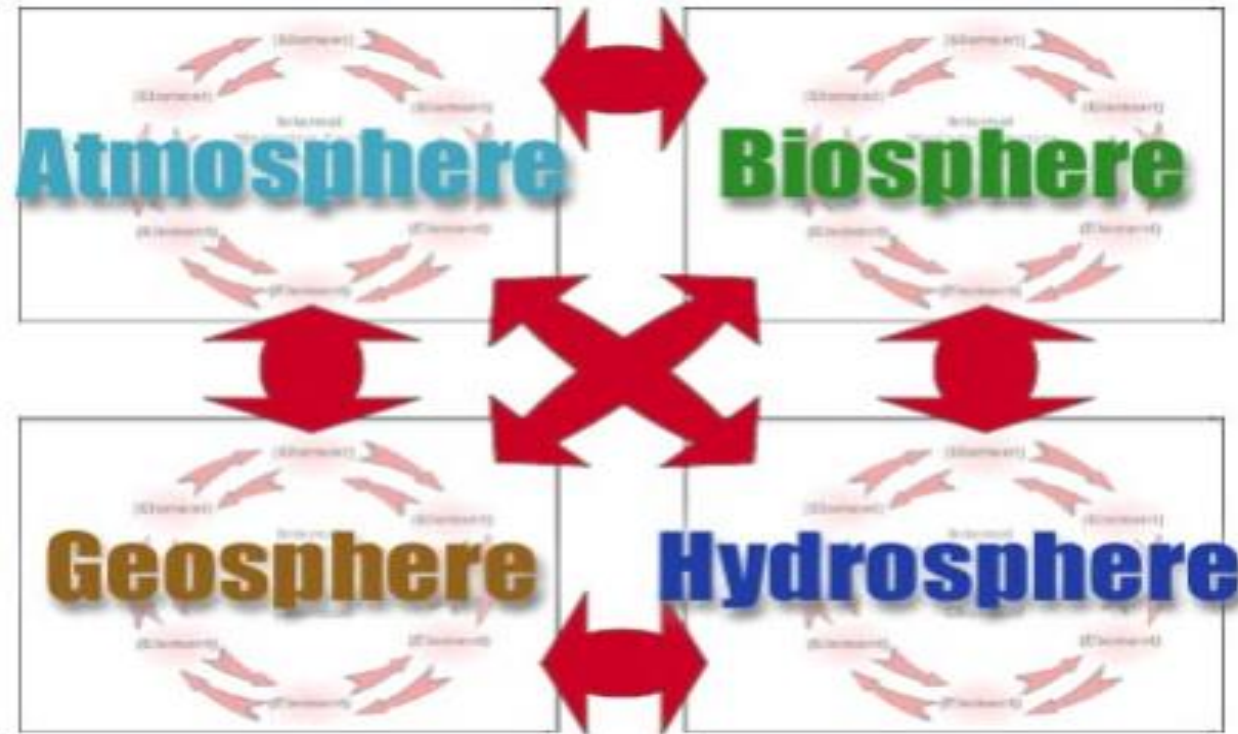


Nitrogen?

- Nitrogen is the most abundant element in the Earth's atmosphere.
- Other major stores of nitrogen include organic matter in soil and the oceans.
- Nitrogen is used by living organisms to produce a number of complex organic molecules like ***amino acids*** (e.g. *Glutamine: $C_5H_{10}N_2O_3$*), ***proteins*** (e.g. *hemoglobin: $C_{2932}H_{4724}N_{828}O_{840}S_8Fe_4$*), and ***nucleic acids*** (such as *DNA & RNA*).
- In plants, much of the nitrogen is used in **chlorophyll molecules** (e.g. *Chlorophyll a: $C_{55}H_{72}MgN_4O_5$*) which are essential for photosynthesis and further growth.

- Although the majority of the air we breathe is N_2 , most of the nitrogen in the atmosphere is unavailable for use by organisms.
- This is because the strong triple bond between the N atoms in N_2 molecules makes it relatively inert, or unreactive.
- In order for plants and animals to be able to use nitrogen, N_2 gas must first be converted to more a chemically available form
such as ammonium (NH_4^+), nitrate (NO_3^-), or organic nitrogen e.g., urea, which has the formula $(NH_2)_2CO$.

- **Nitrogen cycle** refers to the movement of nitrogen between the *atmosphere*, *biosphere*, *hydrosphere* and *geosphere* in different forms



- The five processes that make up the nitrogen cycle are:

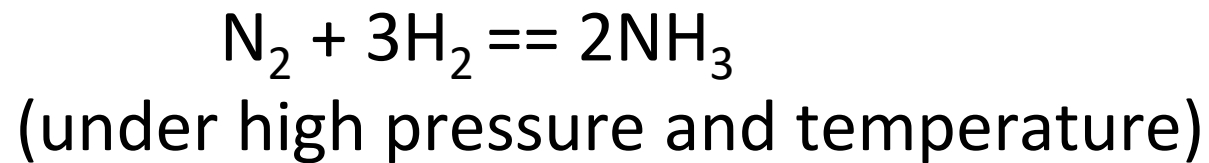
1. *Nitrogen fixation*
2. *Nitrification*
3. *Nitrogen uptake*
4. *Nitrogen Mineralization*
5. *Denitrification*

1. NITROGEN FIXATION

- Nitrogen fixation is the process wherein N_2 is converted to ammonia (NH_3) or nitrogen oxides
- Nitrogen fixation can take place in different ways:
 - ✓ **atmospheric fixation by lightning:** enormous energy of lightning breaks N_2 molecules and enables their atoms to combine with oxygen to form nitrogen oxides



- ✓ **Industrial fixation:** Under great pressure, at a temperature of 600°C, and with the use of a catalyst, atmospheric nitrogen and hydrogen (usually derived from natural gas or petroleum) can be combined to form ammonia (NH₃).



- ✓ **biological fixation** by certain microbes — alone or in a symbiotic relationship with some plants and animals
 - *Certain bacteria, such as Rhizobium, are able to fix nitrogen through metabolic processes*
 - *Some Nitrogen-fixing bacteria often form symbiotic relationships with host plants: This symbiosis is well-known to occur in the legume family of plants (e.g., beans, peas, and clover).*
 - *There are also nitrogen-fixing bacteria that exist without plant hosts, known as free-living nitrogen fixers: e.g. Cyanobacteria*

the chemical reaction of biological N₂-fixation can be written as;



✓ **combustion engine** can also fix N₂,



2. NITRIFICATION

- Nitrification is the biological oxidation of ammonia with oxygen into nitrite (NO_2^-) followed by the oxidation of these nitrites into nitrates (NO_3^-):



- Bacteria of the genus **Nitrosomonas** oxidize NH_4^+ to **nitrites** (NO_2^-).
- Bacteria of the genus **Nitrobacter** oxidize the nitrites to **nitrates** (NO_3^-).

3. NITROGEN UPTAKE

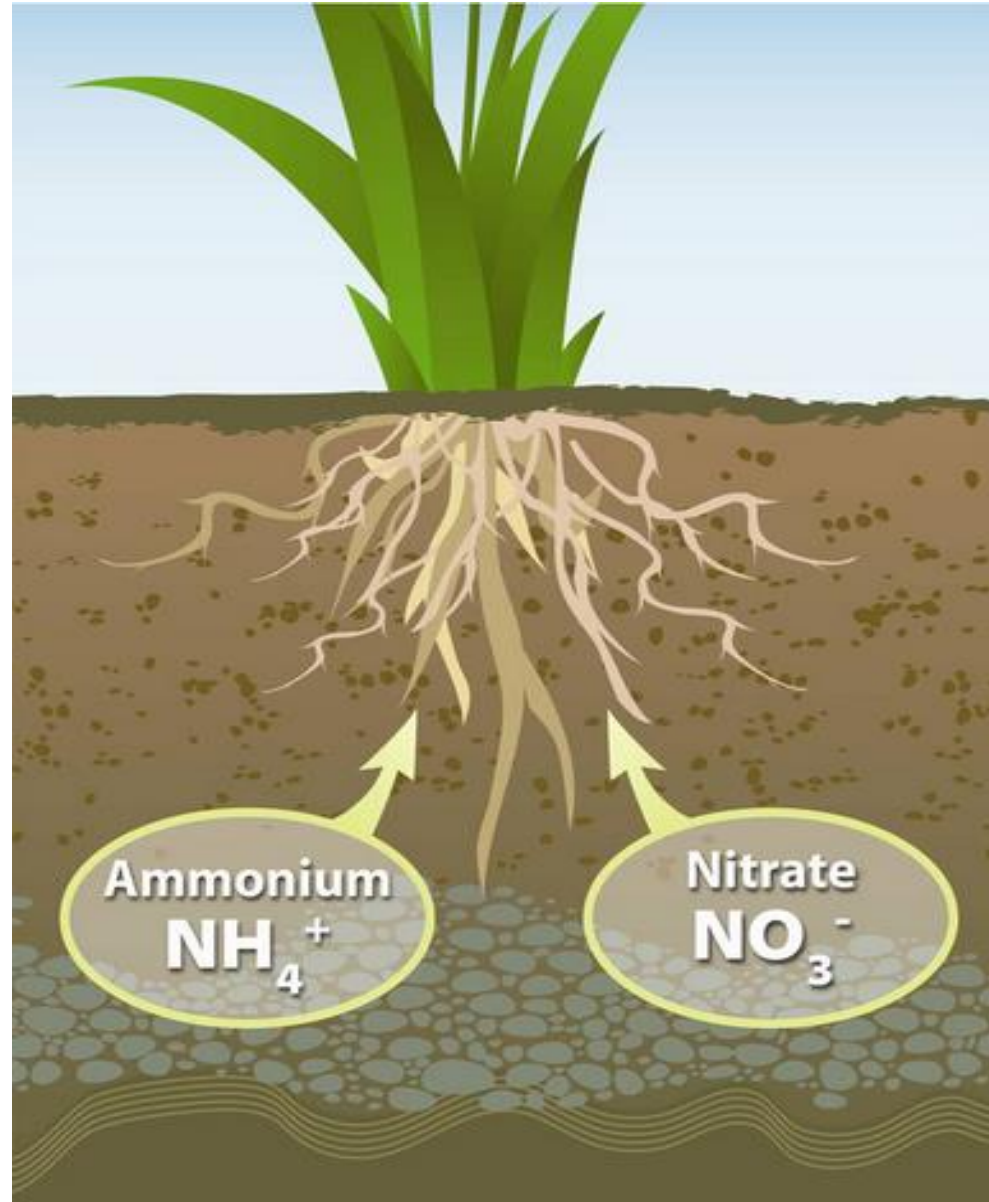
(ASSIMILATION)

- The ammonium (NH_4^+) or Nitrate (NO_3^-) is usually quickly taken up by a host plant, the bacteria itself, or another soil organism and incorporated into proteins and other organic nitrogen compounds, like DNA.

NH_4^+ or NO_3^- to **organic N**

- When organisms nearer the top of the food chain (like us!) eat, we are taking up nitrogen that has been fixed initially by nitrogen-fixing bacteria.

- Plants take up nitrogen either as ammonia, nitrate, or nitrite via their roots.
- Animals get their nitrogen from the food they consume, passing the nitrogen farther and farther up the food chain.



4. NITROGEN MINERALIZATION (DECAY)

- Nitrogen mineralization is a process by which organic Nitrogen is converted back to inorganic nitrogen, which is ammonium ion :

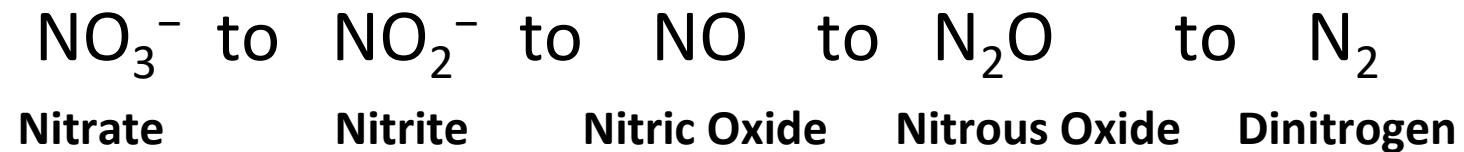
Organic N to NH_4^+

- When organisms die, decomposers consume the organic matter and lead to the process of decomposition.
- During this process, a significant amount of the nitrogen contained within the dead organism is converted to ammonium.

Then, NH_4^+ gets converted to NO_3^- by Nitrification

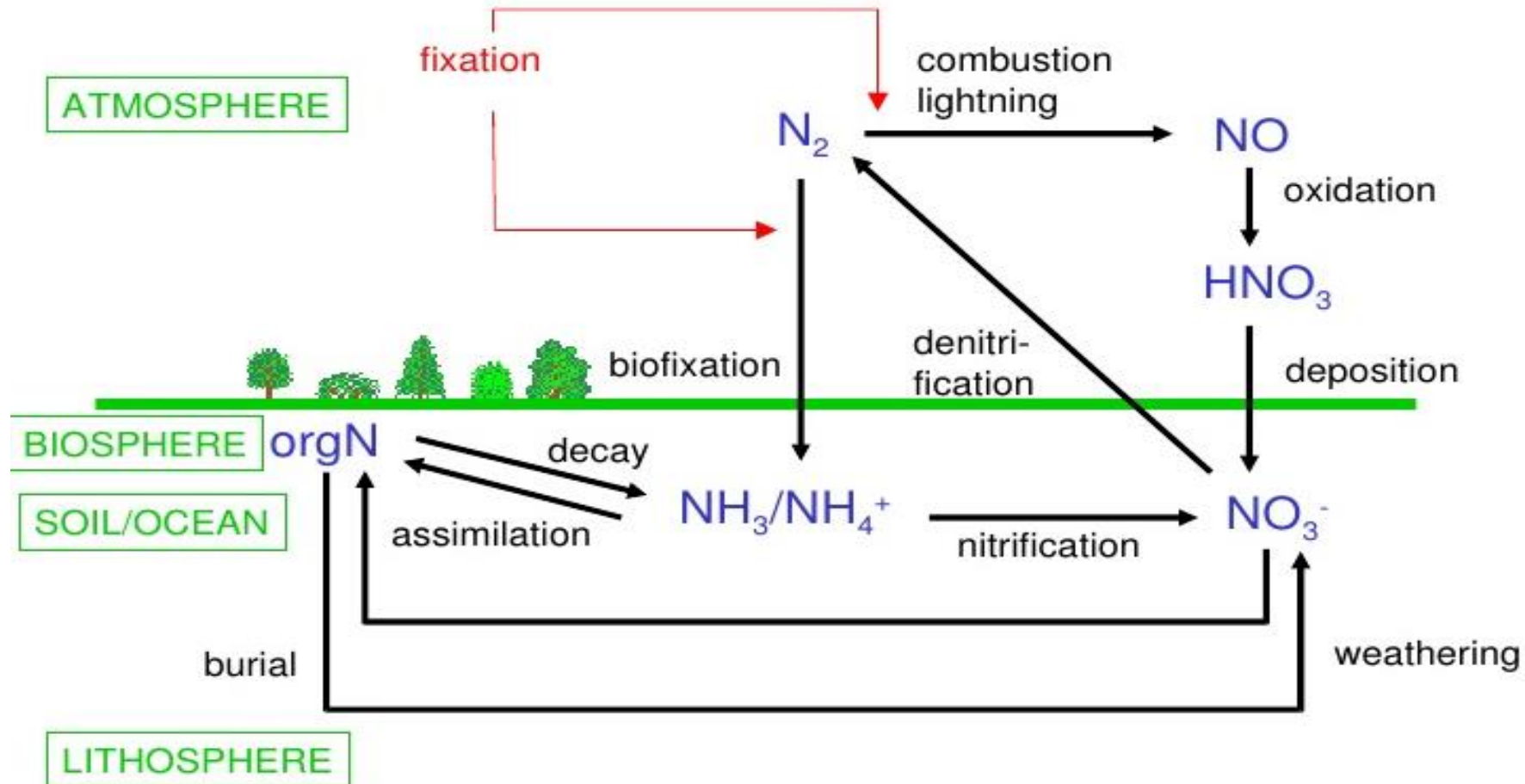
5. DENITRIFICATION

- Denitrification is an anaerobic process that is carried out by denitrifying bacteria, which convert nitrate to dinitrogen (N_2) in the following sequence:



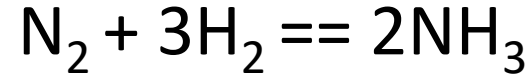
- Once converted to dinitrogen, nitrogen is rapidly lost to the atmosphere completing the nitrogen cycle.

THE NITROGEN CYCLE: MAJOR PROCESSES



Q. Show how anthropogenic perturbation of the nitrogen cycle is taking place?

production of fertilizer



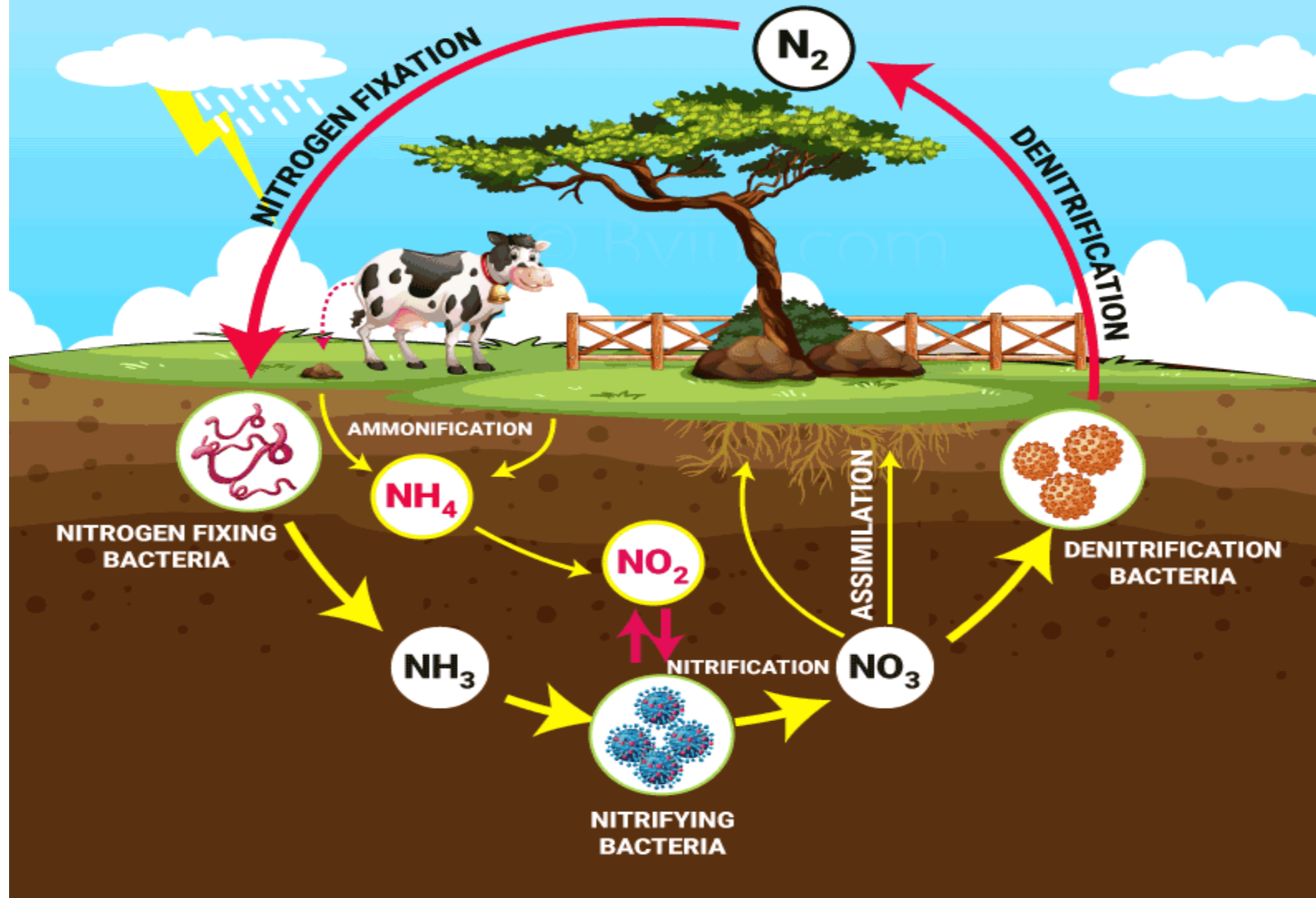
NO emission from combustion of fossil fuel



Use of N_2 -fixing crops in agriculture



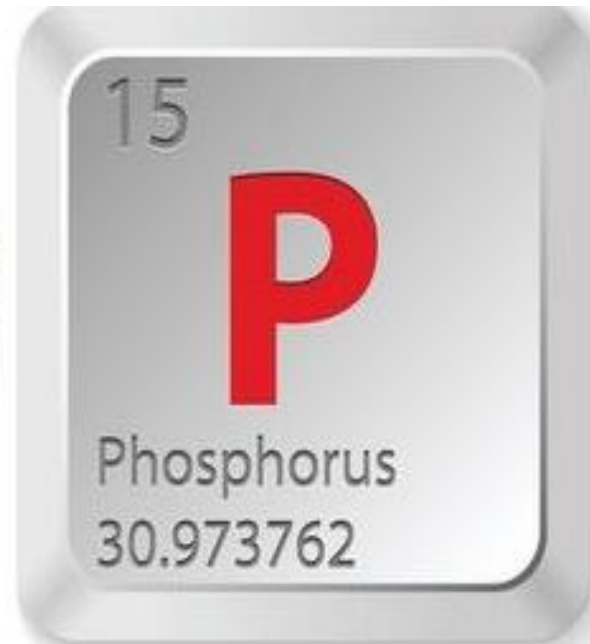
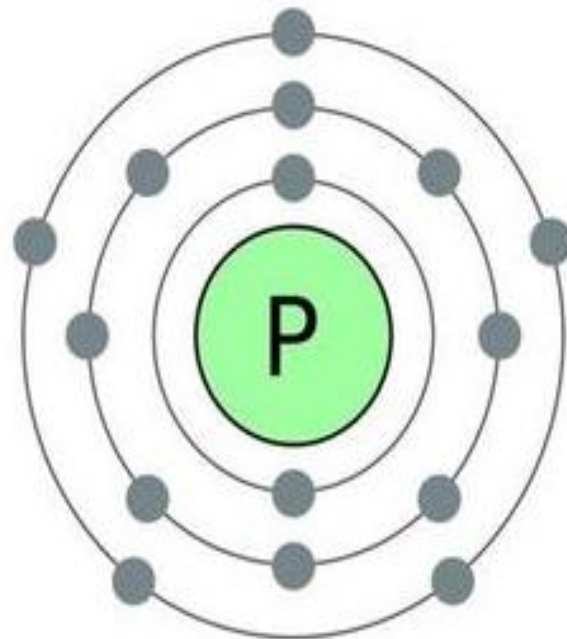
NITROGEN CYCLE



Importance of the Nitrogen Cycle in Meteorology

- ❑ Nitrogen is the dominant gas (78%), influencing air density, pressure, and molecular weight of dry air.
- ❑ Nitrogen oxides (NO_x) and ammonia (NH_3) interact with water vapor formation of *acid rain* (HNO_3 , NH_4^+ compounds).
- ❑ NO and NO_2 are precursors to ozone (O_3) and photochemical smog, major topics in atmospheric chemistry and air quality forecasting.
- ❑ N_2O (nitrous oxide) is a potent greenhouse gas ($\approx 300\times$ stronger than CO_2) and affects radiative forcing in climate models.
- ❑ Ammonium nitrate (NH_4NO_3) aerosols influence visibility, cloud condensation nuclei (CCN), and radiation balance.

The Phosphorus Cycle



Phosphorous?

- Phosphorous is an essential nutrient found in the macromolecules of humans and other organisms, including DNA
- It is a part of molecules that store energy (ATP and ADP) and of fats of cell membranes.
- As calcium phosphate, it also makes up the supportive components of our bones and teeth.
- Due to its high reactivity Phosphorus is never found as a free element.

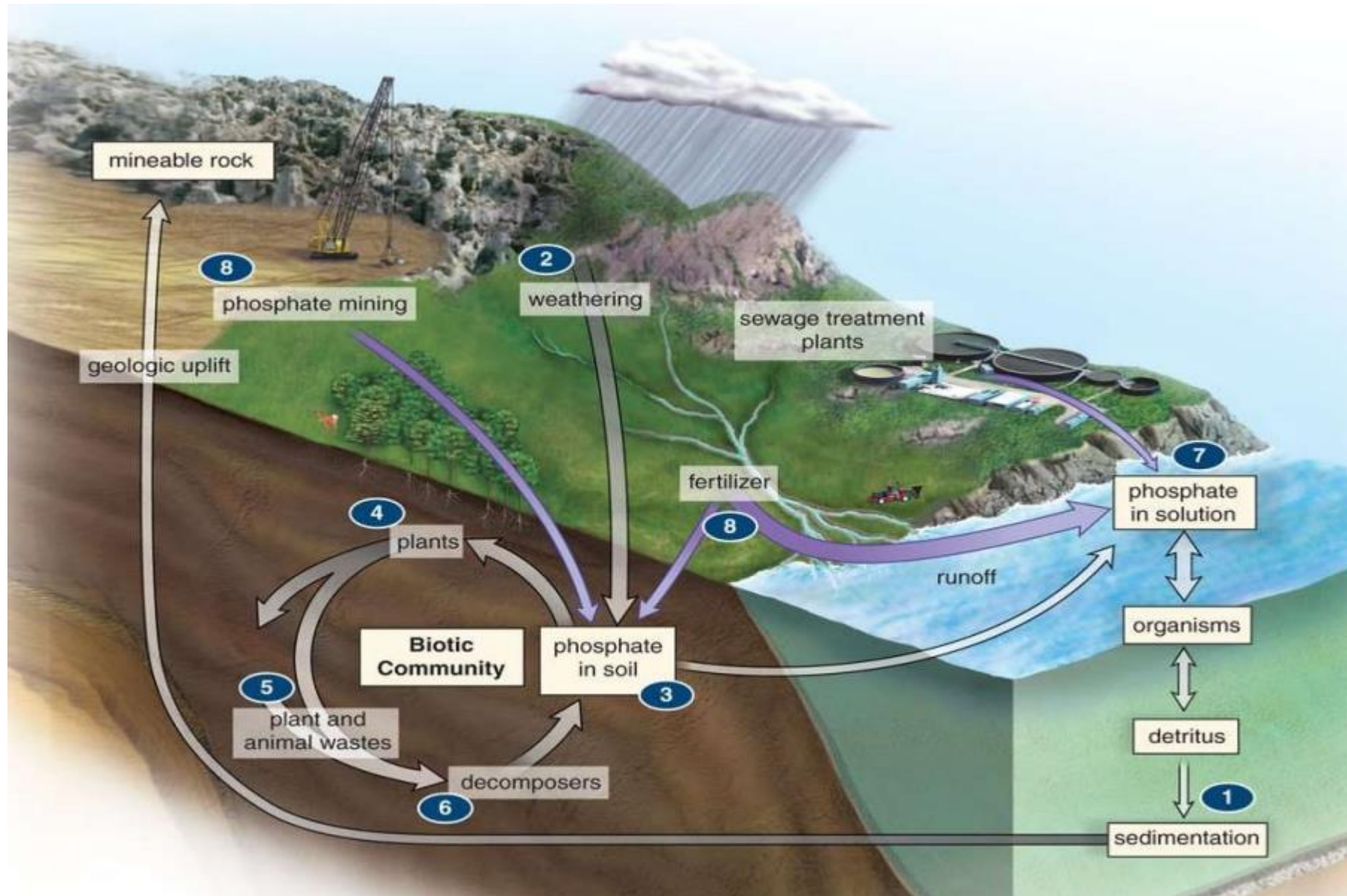
- Most phosphorous in nature exists in the form of phosphate ion (PO_4^{3-})
 - Phosphorous has no real gas phase, unlike other elements such as carbon and nitrogen.
-
- Thus the Phosphorus cycle, unlike those of Carbon and Nitrogen cycles, lacks an atmospheric component.
 - In rock formations and ocean sediments Phosphorus is most commonly found as phosphate salts.

- **The phosphorous cycle** describes the movement of phosphorous through the *lithosphere*, *hydrosphere* and *biosphere*.
- Phosphate compounds are found in sedimentary rocks, and as the rocks weather—wear down over long time periods—the phosphorous they contain slowly leaches into surface water and soils.
- Weathering and erosion of this rocks gradually releases phosphorus as phosphate ions (PO_4^{3-}), which are soluble in water.
- Also, phosphate mining and use of fertilizers pumps phosphorus as phosphate ions to the environment.

- The phosphate ions then accumulate in soil
- Land plants which need phosphate as a fertilizer or nutrient exploit some of it.
- Some of the Phosphate is then transferred to animals that eat the plants.
- When plants and animals excrete wastes or die, the phosphate is then released and returned to the soil for reuse.
- Phosphorous-containing compounds may also be carried in surface runoff to rivers, lakes, and oceans, where they are taken up by aquatic organisms.

- When phosphorous-containing compounds from the bodies or wastes of marine organisms sink to the floor of the ocean, they form new sedimentary layers.
- Over long periods of time, phosphorous-containing sedimentary rock may be moved from the ocean to the land by a geological process called uplift.

Diagrammatic representation of the global phosphorous cycle



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Thanks !